

## HUMA, China

WMO: 503530

Lat: 51.7261N

Lon: 126.6389E

Elev: 576

StdP: 14.39

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |       | Humidification DP/MCDB and HR |     |       |       |     |       | Coldest Month WS/MCDB |      |      |      | MCWS/PCWD to 99.6% DB |      | WSF   |
|---------------|------------|-------|-------------------------------|-----|-------|-------|-----|-------|-----------------------|------|------|------|-----------------------|------|-------|
|               |            |       | 99.6%                         |     |       | 99%   |     |       | 0.4%                  |      | 1%   |      |                       |      |       |
|               | 99.6%      | 99%   | DP                            | HR  | MCDB  | DP    | HR  | MCDB  | WS                    | MCDB | WS   | MCDB | MCWS                  | PCWD |       |
| (a)           | (b)        | (c)   | (d)                           | (e) | (f)   | (g)   | (h) | (i)   | (j)                   | (k)  | (l)  | (m)  | (n)                   | (o)  | (p)   |
| 1             | -35.5      | -31.8 | -41.0                         | 0.5 | -35.0 | -37.1 | 0.7 | -31.2 | 19.1                  | -1.6 | 16.9 | 0.0  | 0.6                   | 180  | 0.578 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |      |      |      |      |      | Evaporation WB/MCDB |      |      |      |      |      | MCWS/PCWD to 0.4% DB |      |
|---------------|------------------------|-----------------|------|------|------|------|------|---------------------|------|------|------|------|------|----------------------|------|
|               |                        | 0.4%            |      | 1%   |      | 2%   |      | 0.4%                |      | 1%   |      | 2%   |      |                      |      |
|               |                        | DB              | MCWB | DB   | MCWB | DB   | MCWB | WB                  | MCDB | WB   | MCDB | WB   | MCDB | MCWS                 | PCWD |
| (a)           | (b)                    | (c)             | (d)  | (e)  | (f)  | (g)  | (h)  | (i)                 | (j)  | (k)  | (l)  | (m)  | (n)  | (o)                  | (p)  |
| 7             | 18.8                   | 88.3            | 67.9 | 85.0 | 67.5 | 81.8 | 66.3 | 72.6                | 82.6 | 71.0 | 80.1 | 69.3 | 77.6 | 7.0                  | 160  |

| Dehumidification DP/MCDB and HR |       |      |      |       |      |      |      |      | Enthalpy/MCDB |      |      |      |      |      | Extreme<br>Max WB |
|---------------------------------|-------|------|------|-------|------|------|------|------|---------------|------|------|------|------|------|-------------------|
| 0.4%                            |       |      | 1%   |       |      | 2%   |      |      | 0.4%          |      | 1%   |      | 2%   |      |                   |
| DP                              | HR    | MCDB | DP   | HR    | MCDB | DP   | HR   | MCDB | Enth          | MCDB | Enth | MCDB | Enth | MCDB |                   |
| (a)                             | (b)   | (c)  | (d)  | (e)   | (f)  | (g)  | (h)  | (i)  | (j)           | (k)  | (l)  | (m)  | (n)  | (o)  |                   |
| 69.5                            | 110.7 | 77.1 | 68.0 | 105.0 | 75.6 | 66.4 | 99.2 | 73.6 | 36.8          | 82.9 | 35.3 | 80.1 | 33.9 | 78.0 | 79.3              |

## Extreme Annual Design Conditions

| Extreme Annual WS |      |      | Extreme Annual Temperature |      |                    |     | n-Year Return Period Values of Extreme Temperature |      |            |      |            |       |            |       |    |    |
|-------------------|------|------|----------------------------|------|--------------------|-----|----------------------------------------------------|------|------------|------|------------|-------|------------|-------|----|----|
|                   |      |      | Mean                       |      | Standard Deviation |     | n=5 years                                          |      | n=10 years |      | n=20 years |       | n=50 years |       |    |    |
| 1%                | 2.5% | 5%   | Min                        | Max  | Min                | Max | Min                                                | Max  | Min        | Max  | Min        | Max   | Min        | Max   | DB | WB |
| (a)               | (b)  | (c)  | (e)                        | (f)  | (g)                | (h) | (i)                                                | (j)  | (k)        | (l)  | (m)        | (n)   | (o)        | (p)   |    |    |
| (4) 19.5          | 16.3 | 13.7 | -38.9                      | 95.1 | 5.1                | 3.6 | -42.5                                              | 97.7 | -45.5      | 99.7 | -48.3      | 101.7 | -52.0      | 104.3 |    |    |
| (5)               |      |      | -38.9                      | 75.2 | 5.3                | 1.9 | -42.7                                              | 76.6 | -45.8      | 77.7 | -48.7      | 78.7  | -52.5      | 80.1  |    |    |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec   |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)   |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 31.9   | -11.7 | -2.3  | 16.1  | 37.8  | 53.1  | 65.5  | 70.5  | 65.8  | 52.9  | 34.5  | 8.4   | -10.4 |  |  |
|                                                                        | DBStd       | 30.70  | 9.97  | 9.59  | 11.23 | 8.14  | 7.49  | 6.43  | 5.18  | 5.78  | 7.74  | 9.00  | 12.79 | 9.77  |  |  |
|                                                                        | HDD50       | 8519   | 1912  | 1463  | 1051  | 376   | 51    | 1     | 0     | 1     | 57    | 486   | 1248  | 1873  |  |  |
|                                                                        | HDD65       | 12457  | 2377  | 1883  | 1516  | 816   | 376   | 69    | 13    | 59    | 366   | 946   | 1698  | 2338  |  |  |
|                                                                        | CDD50       | 1897   | 0     | 0     | 0     | 10    | 148   | 466   | 634   | 489   | 145   | 5     | 0     | 0     |  |  |
|                                                                        | CDD65       | 360    | 0     | 0     | 0     | 0     | 8     | 84    | 182   | 82    | 4     | 0     | 0     | 0     |  |  |
|                                                                        | CDH74       | 3588   | 0     | 0     | 0     | 14    | 199   | 1027  | 1542  | 723   | 81    | 2     | 0     | 0     |  |  |
|                                                                        | CDH80       | 1096   | 0     | 0     | 0     | 2     | 49    | 352   | 504   | 182   | 7     | 0     | 0     | 0     |  |  |
|                                                                        | WSAvg       | 4.6    | 3.1   | 3.6   | 5.3   | 6.6   | 6.3   | 4.8   | 4.4   | 4.1   | 4.6   | 5.1   | 3.8   | 3.0   |  |  |
|                                                                        | PrecAvg     | 18.4   | 0.2   | 0.1   | 0.4   | 1.1   | 1.5   | 2.8   | 4.5   | 4.3   | 2.2   | 0.8   | 0.4   | 0.3   |  |  |
| (15) Precipitation                                                     | PrecMax     | 24.2   | 0.5   | 0.4   | 1.0   | 3.7   | 3.3   | 6.8   | 7.4   | 9.8   | 7.3   | 1.9   | 1.8   | 0.7   |  |  |
|                                                                        | PrecMin     | 13.3   | 0.0   | 0.0   | 0.0   | 0.1   | 0.4   | 0.6   | 1.8   | 1.7   | 0.5   | 0.2   | 0.1   | 0.0   |  |  |
|                                                                        | PrecStd     | 3.0    | 0.1   | 0.1   | 0.2   | 0.7   | 0.9   | 1.6   | 1.6   | 1.9   | 1.4   | 0.5   | 0.4   | 0.2   |  |  |
|                                                                        | PrecStd     | 3.0    | 0.1   | 0.1   | 0.2   | 0.7   | 0.9   | 1.6   | 1.6   | 1.9   | 1.4   | 0.5   | 0.4   | 0.2   |  |  |
| (19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 0.4%        | DB     | 17.4  | 29.9  | 51.4  | 75.1  | 86.4  | 93.9  | 94.4  | 89.1  | 80.7  | 67.2  | 41.7  | 22.2  |  |  |
|                                                                        |             | MCWB   | 14.7  | 24.4  | 38.2  | 50.5  | 59.1  | 67.0  | 72.2  | 68.8  | 63.8  | 50.6  | 34.2  | 19.1  |  |  |
|                                                                        | 2%          | DB     | 11.8  | 23.1  | 44.4  | 65.0  | 79.7  | 87.7  | 88.8  | 84.8  | 76.3  | 59.1  | 35.3  | 14.7  |  |  |
|                                                                        |             | MCWB   | 9.4   | 19.7  | 34.0  | 46.4  | 56.4  | 65.6  | 70.1  | 69.7  | 61.5  | 46.6  | 30.3  | 12.6  |  |  |
|                                                                        | 5%          | DB     | 7.7   | 18.0  | 38.9  | 58.8  | 74.7  | 84.0  | 85.4  | 81.3  | 71.8  | 53.9  | 30.9  | 8.3   |  |  |
|                                                                        |             | MCWB   | 5.9   | 15.0  | 30.9  | 43.5  | 54.1  | 64.7  | 69.3  | 68.1  | 59.1  | 43.3  | 27.6  | 6.6   |  |  |
|                                                                        | 10%         | DB     | 3.6   | 13.5  | 34.4  | 53.9  | 69.8  | 80.0  | 82.1  | 77.9  | 67.4  | 48.9  | 26.7  | 4.1   |  |  |
|                                                                        |             | MCWB   | 2.1   | 11.1  | 28.5  | 40.5  | 52.2  | 63.2  | 68.5  | 66.4  | 56.6  | 40.1  | 23.7  | 2.7   |  |  |
|                                                                        | 0.4%        | WB     | 14.4  | 24.6  | 39.2  | 51.9  | 61.6  | 72.2  | 75.0  | 73.9  | 67.0  | 52.9  | 34.9  | 19.5  |  |  |
|                                                                        |             | MCDB   | 17.1  | 29.0  | 50.2  | 72.1  | 79.6  | 86.2  | 85.9  | 83.4  | 75.1  | 64.9  | 40.6  | 22.9  |  |  |
| (27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 2%          | WB     | 9.4   | 20.0  | 34.6  | 47.2  | 58.8  | 69.0  | 73.2  | 71.8  | 63.7  | 47.5  | 31.1  | 12.6  |  |  |
|                                                                        |             | MCDB   | 11.6  | 23.3  | 43.1  | 63.0  | 73.7  | 80.5  | 83.6  | 80.2  | 72.3  | 57.2  | 34.7  | 14.8  |  |  |
|                                                                        | 5%          | WB     | 6.0   | 15.1  | 31.5  | 44.2  | 56.3  | 66.9  | 71.8  | 70.1  | 61.2  | 44.1  | 27.7  | 6.7   |  |  |
|                                                                        |             | MCDB   | 7.8   | 17.9  | 38.5  | 57.3  | 70.1  | 78.4  | 81.5  | 78.0  | 69.0  | 52.2  | 31.0  | 8.2   |  |  |
|                                                                        | 10%         | WB     | 2.1   | 11.2  | 28.5  | 41.2  | 53.9  | 65.1  | 70.4  | 68.1  | 58.5  | 40.9  | 23.6  | 2.7   |  |  |
|                                                                        |             | MCDB   | 3.5   | 13.5  | 34.3  | 52.1  | 66.2  | 76.7  | 79.1  | 75.4  | 65.8  | 48.4  | 26.7  | 4.0   |  |  |
|                                                                        | 5% DB       | MDBR   | 19.5  | 24.8  | 24.0  | 22.2  | 23.4  | 22.1  | 18.8  | 20.0  | 22.2  | 19.7  | 18.5  | 17.4  |  |  |
|                                                                        |             | MCDBR  | 21.0  | 28.5  | 26.6  | 32.5  | 33.6  | 29.5  | 25.3  | 25.9  | 28.5  | 28.4  | 21.0  | 19.0  |  |  |
|                                                                        |             | MCWBR  | 19.0  | 24.9  | 18.0  | 18.3  | 14.9  | 11.5  | 9.8   | 12.2  | 14.6  | 16.9  | 16.3  | 17.4  |  |  |
|                                                                        |             | MCDBR  | 20.7  | 28.2  | 25.5  | 29.8  | 27.5  | 24.1  | 20.3  | 21.2  | 24.5  | 25.0  | 20.3  | 18.8  |  |  |
|                                                                        | 5% WB       | MCWBR  | 18.8  | 24.7  | 17.5  | 17.2  | 13.0  | 11.0  | 9.5   | 11.2  | 13.6  | 15.9  | 15.9  | 17.2  |  |  |
|                                                                        |             | MCWBR  | 18.8  | 24.7  | 17.5  | 17.2  | 13.0  | 11.0  | 9.5   | 11.2  | 13.6  | 15.9  | 15.9  | 17.2  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        |        | 0.265 | 0.298 | 0.375 | 0.541 | 0.613 | 0.515 | 0.481 | 0.421 | 0.385 | 0.371 | 0.310 | 0.272 |  |  |
|                                                                        | taud        |        | 2.160 | 2.089 | 1.934 | 1.699 | 1.633 | 1.963 | 2.119 | 2.279 | 2.301 | 2.179 | 2.153 | 2.107 |  |  |
|                                                                        | Ebn at Noon |        | 233   | 255   | 252   | 220   | 211   | 239   | 245   | 255   | 251   | 224   | 208   | 203   |  |  |
|                                                                        | Edh at Noon |        | 23    | 33    | 47    | 68    | 76    | 55    | 47    | 38    | 33    | 30    | 23    | 20    |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      |        | 473   | 849   | 1381  | 1625  | 1708  | 1910  | 1723  | 1510  | 1189  | 781   | 491   | 364   |  |  |
|                                                                        | RadStd      |        | 16    | 23    | 65    | 172   | 133   | 211   | 153   | 103   | 94    | 48    | 23    | 18    |  |  |

## Historical Trends

|                             | DBAvg | Heating |     |        | Cooling |       |       | Degree-Days |       |       |       |
|-----------------------------|-------|---------|-----|--------|---------|-------|-------|-------------|-------|-------|-------|
|                             |       | 99% DB  |     | 99% DP | 1% DB   | 1% WB | 1% DP | HDD50       | HDD65 | CDD50 | CDD65 |
|                             |       | (d)     | (e) | (f)    | (g)     | (h)   | (i)   | (j)         | (k)   | (l)   | (m)   |
| (46) Station Only           | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |
| (47) Regional (8 neighbors) | N/A   | N/A     | N/A | N/A    | N/A     | N/A   | N/A   | N/A         | N/A   | N/A   | N/A   |

Nomenclature: See separate page